

Comparing ML and LLM Approaches for PTSD Risk Prediction: A Failure Analysis Study

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Spring 2026

Abstract

This project investigates whether a large language model (LLM) can predict depression and PTSD risk from structured health survey data as effectively as traditional machine learning models. Using the NHANES 2017-2018 dataset, I trained three ML baselines — Logistic Regression, Random Forest, and XGBoost — and compared them against Llama 3 70B prompted in eight different ways. Beyond simple performance comparison, I analyzed where and why the LLM fails, evaluated the faithfulness of its clinical explanations against SHAP-derived feature importance, and tested whether a structured debate between two LLM instances could resolve uncertainty on borderline cases. The results show that ML models significantly outperform the LLM on tabular clinical data, but the more interesting finding is that LLM failures are not random — they cluster specifically on borderline PHQ scores of 8-9, just below the classification threshold. This pattern persists across all prompting strategies including few-shot, SHAP-guided, and debate-based approaches, suggesting a fundamental limitation in how LLMs reason about hard numerical thresholds in clinical settings.

1. Introduction

Mental health prediction has always been tricky — unlike imaging where models clearly win, survey data is messier and the labels are rarely perfect. Unlike imaging tasks where deep learning has clearly won, structured survey data presents a different challenge — the patterns are often subtle, the labels are imperfect proxies, and the stakes of getting it wrong are high.

Most work in this area uses traditional ML models trained on tabular features and evaluates them on standard metrics. What gets less attention is whether large language models —

which have absorbed enormous amounts of medical literature — can make reasonable predictions from the same data without any task-specific training. And if they can, do they reason about it correctly?

This project tries to answer those questions using NHANES 2017-2018 survey data. The dataset includes PHQ-9 depression screening responses, demographics, sleep, and alcohol data for over 5000 Americans. While the project is framed as PTSD risk prediction, the label is derived from PHQ-9 depression scores due to the absence of direct PTSD diagnoses in NHANES. PHQ-9 is used as a validated clinical proxy for depression severity, and this limitation is acknowledged throughout. I used PHQ-9 score to construct a binary risk label, trained three ML models as baselines, and then ran a series of experiments prompting Llama 3 70B to make the same predictions from text descriptions of each patient.

The interesting part of this project is not the performance comparison — it was always likely that trained ML models would win. The interesting part is the failure analysis: finding that every single LLM error clusters on borderline PHQ scores of 8-9, and that this cannot be fixed through better prompting, few-shot examples, SHAP-guided context, or even structured debate between two LLM instances.

2. Related Work

Several recent papers have explored LLMs for mental health assessment, though most focus on text-based inputs rather than structured tabular survey data — which is the gap this project addresses.

Weber et al. [1] fine-tuned a German BERT-based model on clinical interview transcripts to predict MADRS depression scores. Their fine-tuned model achieved 79-88% accuracy, closely matching clinician ratings, with a 75% reduction in errors over the untrained baseline. This work shows that fine-tuning on domain-specific text works well — but it requires actual clinical interview transcripts, not the structured tabular data that most health surveys produce.

Lee et al. [2] focused on interpretable depression assessment using LLMs, finding that LLMs can generate clinically meaningful explanations alongside predictions, but the faithfulness of those explanations to underlying data patterns varied. Their work directly motivated the interpretability and faithfulness evaluation in this project.

Lundberg and Lee [3] introduced SHAP (SHapley Additive exPlanations), now the standard tool for explaining tree-based model predictions. I used SHAP to generate per-patient feature importance scores as ground truth for evaluating LLM explanation quality.

Unlike these prior works which focus primarily on text-based inputs or model interpretability in isolation, this study directly compares zero-shot LLM prompting against trained tabular classifiers on structured clinical data, and evaluates not just prediction accuracy but reasoning faithfulness and robustness to adversarial debate.

3. Methodology

3.1 Dataset

I used the NHANES 2017-2018 dataset, a nationally representative health survey conducted by the CDC. The 2017-2018 cycle was chosen because it is the last complete pre-COVID cycle available as a standalone dataset — later cycles were merged due to COVID disruptions. Four files were downloaded and merged on the patient ID (SEQN): the PHQ-9 depression screener (DPQ), demographics (DEMO), alcohol use (ALQ), and sleep data (SLQ). After merging and cleaning, the final dataset had 5065 patients and 22 features.

3.2 Label Construction

NHANES does not include a direct PTSD diagnosis label. I constructed a binary risk label using the PHQ-9 total score, calculated by summing responses to nine depression symptom questions (each scored 0-3). Patients with PHQ-9 total ≥ 10 were labeled high risk (1), and those below were labeled low risk (0). This threshold is the standard clinical cutoff for moderate depression [4]. The resulting dataset was imbalanced: 8.3% high risk and 91.7% low risk, which motivated the use of F1 macro and ROC-AUC as primary metrics rather than accuracy.

3.3 Workflow

Figure 1 shows the overall project pipeline from data collection through analysis.

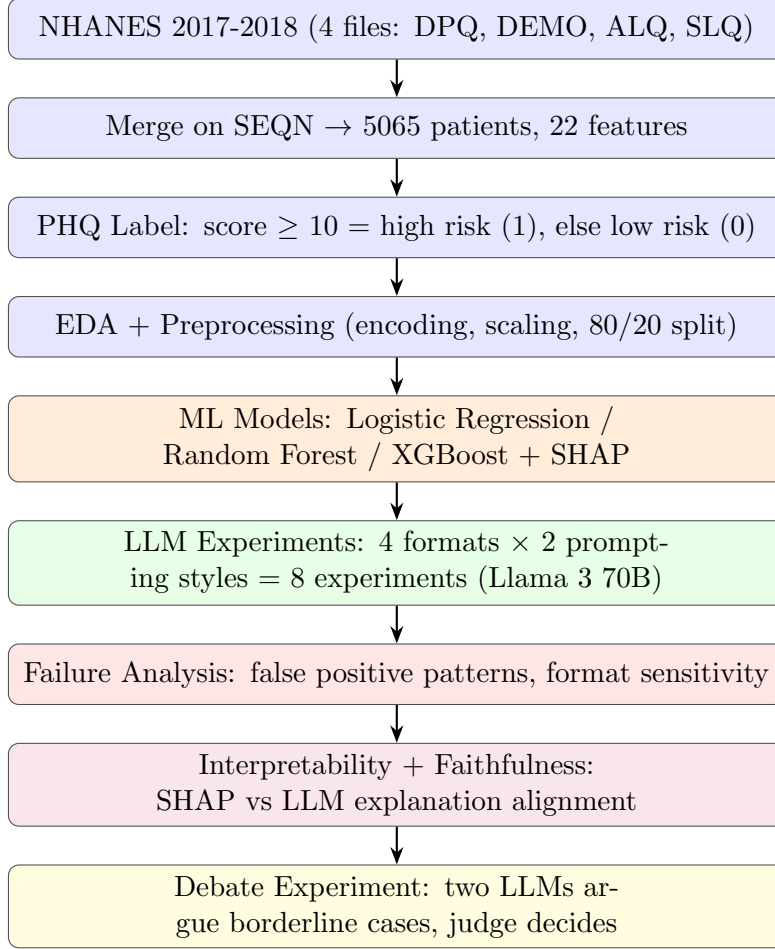


Figure 1: Project workflow from data collection to analysis.

3.4 Preprocessing

PHQ_total was dropped before training to prevent data leakage — the label is directly derived from it, so keeping it would let the model essentially look up the answer. The patient ID (SEQN) was also dropped. Categorical columns (gender, race, education, marital status) were one-hot encoded using `pd.get_dummies`. Continuous features (age, income ratio, sleep hours) were standardized using `StandardScaler` for Logistic Regression only — tree-based models do not require scaling. The data was split 80/20 with stratification to preserve the class ratio in both sets.

3.5 ML Models

Three models were trained using scikit-learn and XGBoost. All models used `class_weight='balanced'` to handle the class imbalance. Logistic Regression served as the linear baseline. Random Forest used 100 estimators. XGBoost used default parameters with `scale_pos_weight` adjusted for imbalance. After training, SHAP TreeExplainer was run on XGBoost to generate per-patient feature importance scores, identifying the top-3 most influential features for each

test patient.

3.6 LLM Experiments

Llama 3.3 70B was accessed via the Groq API (free tier). Due to rate limits, experiments were run on a random sample of 200 test patients. Four serialization formats were designed to convert tabular rows into text prompts:

- **Format 1 (Basic):** Key-value pairs with direct question
- **Format 2 (Sentence):** Natural language patient description
- **Format 3 (Clinical):** Compact clinical notation
- **Format 4 (SHAP-guided):** Highlights top-3 SHAP features per patient — the novel addition, using XGBoost’s learned feature importance to guide LLM attention

Each format was tested with zero-shot and few-shot prompting (5 labeled examples), giving 8 experiments total. Responses were parsed with regex to extract binary predictions.

3.7 Interpretability and Faithfulness Evaluation

For the 15 high-risk patients in the sample, I prompted the LLM to both predict and explain its reasoning in 2-3 sentences. Explanations were compared against SHAP top-3 features using keyword matching. A faithfulness score was calculated as the proportion of SHAP-important features mentioned in the explanation. Cases where the LLM predicted correctly but had faithfulness < 0.5 were flagged as “right for wrong reasons” — a clinical safety concern.

3.8 Debate Experiment

The 12 borderline patients (PHQ 8-9) that failed consistently across all prompt formats were subjected to a structured debate. Two separate LLM calls were made per patient — one arguing for high risk, one arguing for low risk. A third LLM call acted as judge. This tested whether adversarial reasoning could resolve uncertainty on threshold cases.

4. Results

4.1 Class Distribution

Figure 2 shows the class imbalance and PHQ score distribution. The dataset is heavily skewed toward low risk patients — 91.7% low risk vs 8.3% high risk. The right-skewed PHQ histogram confirms that most survey respondents report minimal depression symptoms, which is expected for a general population sample. This imbalance is why accuracy alone would be misleading as a metric.

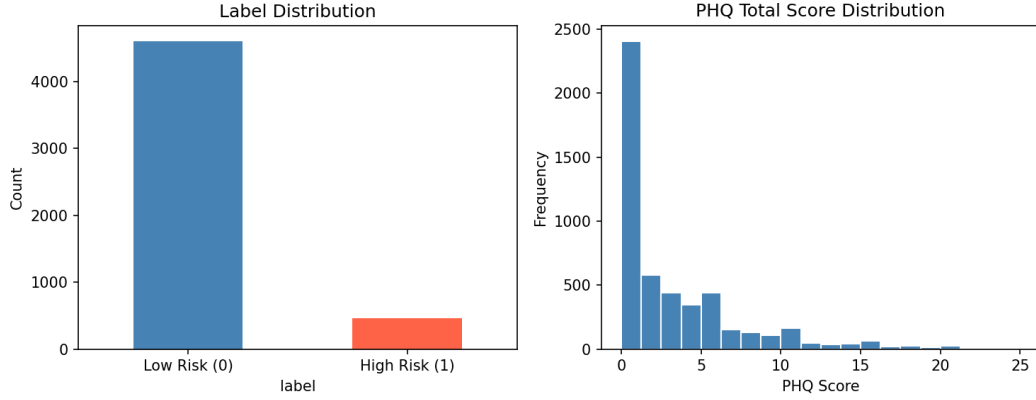


Figure 2: Label distribution (left) and PHQ total score histogram (right). The heavy class imbalance motivated use of F1 macro and AUC rather than accuracy.

4.2 EDA Findings

Figure 3 shows average PHQ question scores by label. High risk patients score consistently higher across all nine questions, with the largest gaps in DPQ040 (feeling down or hopeless) and DPQ030 (sleep problems). This confirms that depression symptoms are the primary signal in the data, and that demographic and lifestyle features play a much weaker role. I found this interesting because it means the classification problem is essentially dominated by the PHQ questions, which has implications for both the ML models and the LLM experiments.

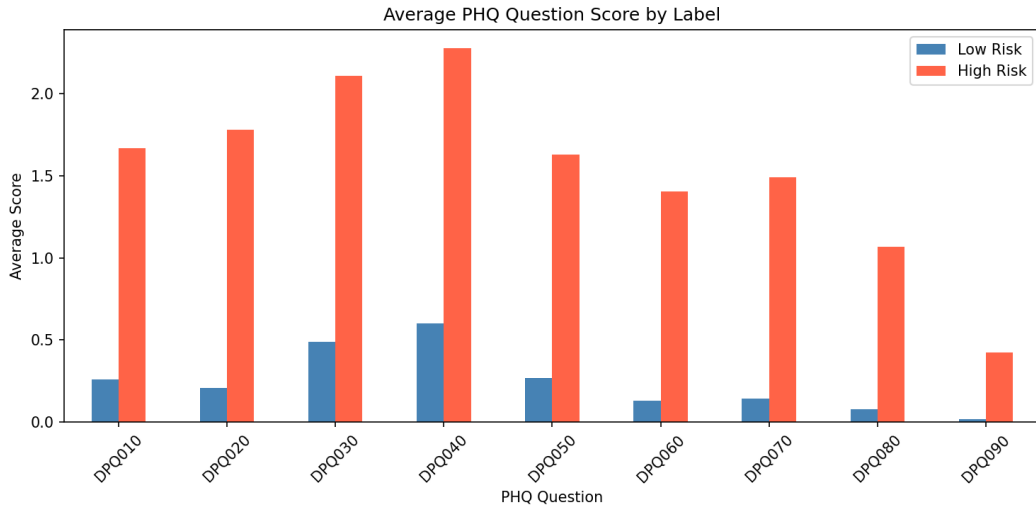


Figure 3: Average PHQ question scores by label. The consistent gap across all nine items confirms PHQ dominates the signal. DPQ040 and DPQ030 show the largest separation.

4.3 ML Model Performance

Table 1 shows performance on the full test set (n=1013). All three models performed very strongly. Logistic Regression achieved the highest F1 (0.969) and an AUC of 1.0. This

initially looks suspicious, but it makes sense on closer inspection: the PHQ questions are used both as features and to construct the label. Because the label is a threshold function of the PHQ sum, and those individual questions are included as features, the model is essentially learning a near-direct mapping. This is not data leakage in the traditional sense because we dropped PHQ_total before training, but the underlying signal is still very strong. It confirms the classification task is nearly linearly separable when PHQ features are present.

Table 1: ML Model Performance on Full Test Set (n=1013)

Model	Accuracy	F1 Macro	AUC
Logistic Regression	0.989	0.969	1.000
XGBoost	0.979	0.940	0.998
Random Forest	0.974	0.912	0.995

4.4 LLM Performance

Table 2 shows results for the three reliable LLM experiments (200/200 valid predictions). Complex formats (F3 clinical, F4 SHAP-guided) produced unparseable verbose responses for many patients — the LLM would explain its reasoning rather than output a clean 0 or 1. This is itself an interesting finding: more technically precise prompts made the LLM less reliable, not more.

Table 2: LLM Performance on Sample (n=200, reliable formats only)

Experiment	Accuracy	F1 Macro	Valid
F1 Zero-shot	0.940	0.840	200/200
F1 Few-shot	0.935	0.831	200/200
F2 Zero-shot	0.935	0.831	200/200

Figure 4 compares F1 scores across all reliable models. The best LLM achieves 0.84 F1, approximately 0.10 below XGBoost. Considering the LLM had zero training on this data, 0.84 is actually surprisingly competitive — but the gap is real and consistent.

4.5 Failure Analysis

The LLM (F1 zero-shot) produced 12 false positives and 0 false negatives. Figure 5 reveals that every false positive had a PHQ score of 8 or 9 — just below the threshold of 10. The age distribution shows no clustering, confirming demographics are not the cause. Nine of these patients failed consistently across all three reliable formats, meaning the failure pattern is not about prompt design — it is about the specific PHQ score range. The LLM appears to treat PHQ=9 as clinically concerning (which is medically reasonable) but our label says low risk because $9 < 10$.

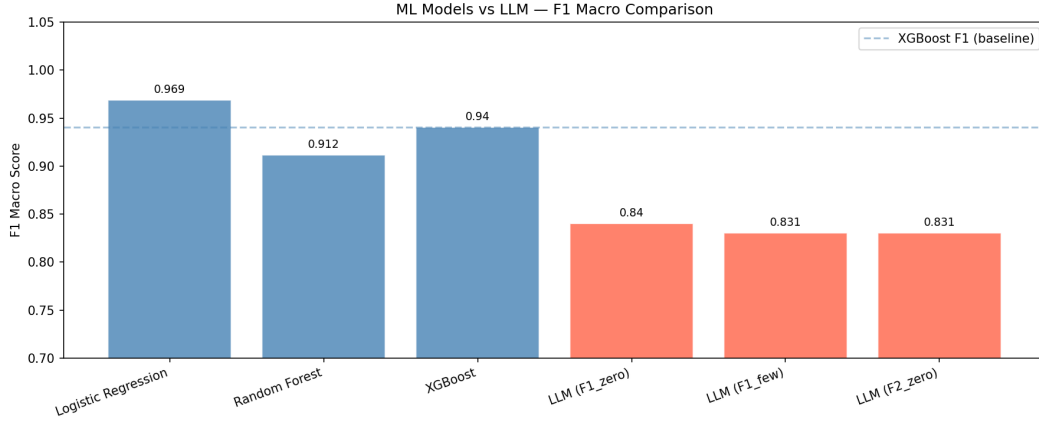


Figure 4: F1 Macro comparison between ML models and reliable LLM experiments. The ~ 0.10 gap between XGBoost and the best LLM is consistent across all three reliable formats.

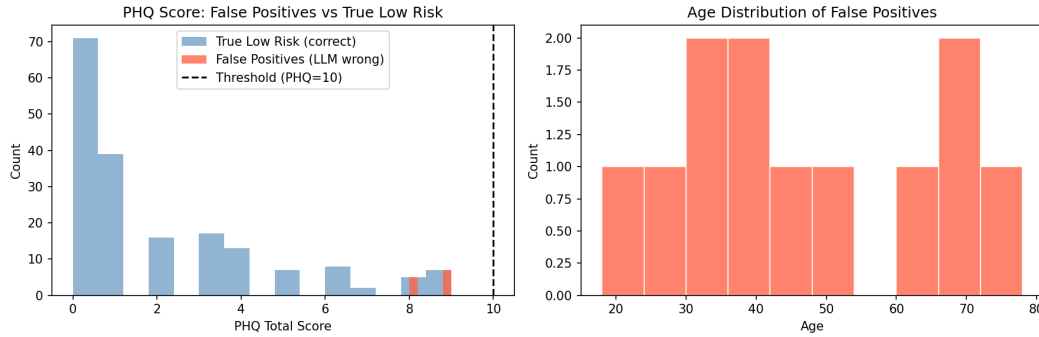


Figure 5: PHQ score distribution of false positives (left) and age distribution (right). Every LLM error falls in the PHQ 8-9 range, while age shows no pattern — confirming the failure is threshold-driven, not demographic.

4.6 Interpretability and Faithfulness

On the 15 high-risk patients, the LLM achieved 60% accuracy overall. Breaking this down by severity revealed a clear pattern: 100% accuracy on patients with $\text{PHQ} > 12$, but only 40% on patients with $\text{PHQ} 10-12$. This confirms that the LLM handles clear cases well but struggles at the boundary. Average faithfulness score was 0.69 — the LLM mentioned approximately 2 out of 3 SHAP-important features in its explanations. Two patients (13%) were “right for wrong reasons”: correct prediction but low faithfulness. This is a clinical safety concern because a doctor trusting the LLM’s explanation might focus on the wrong symptoms.

4.7 Debate Experiment

The structured two-LLM debate on 12 borderline patients produced 0/12 correct verdicts — identical to single-prompt performance. In every case the judge sided with the high-risk argument, even after hearing a coherent low-risk counterargument. One surprising

result here was that the low-risk arguments were actually well-reasoned — but the judge consistently overrode them with clinical conservatism. This suggests the LLM’s tendency to flag borderline cases as high risk is deeply embedded and cannot be resolved through adversarial prompting alone.

5. Limitations

This project has several limitations worth stating clearly. First, the label is derived from PHQ-9 depression scores, not a clinical PTSD diagnosis — PHQ-9 measures depression severity and the two conditions, while related, are clinically distinct. Second, LLM experiments were run on a sample of 200 patients rather than the full test set of 1013, due to Groq API rate limits. Third, no fine-tuning was performed on the LLM — all experiments used zero-shot or few-shot prompting with a general-purpose model. A medical-specific LLM or a fine-tuned model would likely perform differently. Fourth, converting tabular rows to text loses some structural information — the serialization process is imperfect and different formats produced inconsistent LLM behavior. Finally, the faithfulness evaluation uses keyword matching which is a rough proxy for true semantic alignment between LLM explanations and SHAP values.

6. Conclusion

I started this project curious about one thing: can an LLM that has never seen my data still predict PTSD risk just from reading a patient description? The answer turned out to be more nuanced than I expected.

Traditional ML models significantly outperform zero-shot LLM prompting on tabular clinical data, with XGBoost achieving $F1=0.940$ compared to the best LLM at $F1=0.840$. However, the performance gap is not the most interesting result.

But the number I kept coming back to wasn’t the F1 score — it was the PHQ scores of every patient the LLM got wrong: 8, 9, 8, 9, 9, 9. Every single one just below the threshold. They cluster specifically on borderline PHQ cases (scores 8-9), persist across all prompting strategies including few-shot, SHAP-guided, and debate-based approaches, and reflect a fundamental property of how LLMs reason: they apply clinical conservatism rather than strict threshold logic. A patient with $PHQ=9$ is “clinically concerning” by general medical knowledge, but our label says low risk because $9 < 10$. The LLM cannot learn this distinction without labeled examples.

The faithfulness evaluation added another dimension: even when the LLM predicts correctly, its explanations sometimes cite different features than SHAP identifies as important. This “right for wrong reasons” pattern is a clinical safety concern that goes beyond accuracy metrics.

For future work, fine-tuning a smaller LLM on labeled NHANES data could teach it the hard PHQ threshold. It would also be worth testing on actual PTSD diagnosis data rather

than a PHQ proxy, and exploring whether LLMs perform differently on narrative clinical notes versus structured survey responses.

References

- [1] Weber, S., Deperrois, N., Heun, R., et al. Using a fine-tuned large language model for symptom-based depression evaluation. *npj Digital Medicine*, 8:598, 2025.
- [2] Lee, J.J., Han, J., and Woo, C.W. Interpretable depression assessment using a large language model. *PLOS Digital Health*, 5(2):e0001205, 2026.
- [3] Lundberg, S.M. and Lee, S.I. A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30, 2017.
- [4] Kroenke, K., Spitzer, R.L., and Williams, J.B. The PHQ-9: validity of a brief depression severity measure. *Journal of General Internal Medicine*, 16(9):606–613, 2001.